EE2025: Engineering Electromagnetics

Tutorial 2: Maxwell's equations

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1. Show that charge density ρ_v satisfies,

$$\frac{\partial \rho_v}{\partial t} + \frac{\sigma}{\epsilon}\rho_v = 0$$

Is this true for all media?

Solution:

From Gauss's law, we have,

$$\nabla \cdot \mathbf{D} = \rho_v,$$

where $\mathbf{D} = \epsilon \mathbf{E}$

$$\implies \nabla \cdot \mathbf{E} = \frac{\rho_v}{\epsilon}$$

From continuity equation and Ohm's law, we can write,

$$\nabla \cdot \mathbf{J} = \nabla \cdot \sigma \mathbf{E} = \sigma \nabla \cdot \mathbf{E} = -\frac{\partial \rho_v}{\partial t}$$

$$\implies \sigma \frac{\rho_v}{\epsilon} = -\frac{\partial \rho_v}{\partial t}$$

Or we can write,

$$\frac{\partial \rho_v}{\partial t} + \frac{\sigma}{\epsilon}\rho_v = 0$$

The above differential equation holds true for a linear, homogeneous, isotropic medium. This is because $\mathbf{J} = \sigma\rho$ (Ohm's Law) would be valid only in such a medium.

2. A slab of perfect dielectric material ($\epsilon_r = 2$) is placed in a microwave oven. The oven produces an electric field (as well as a magnetic field). Assume that the electric field intensity is uniform in space and sinusoidal in time and is in the direction perpendicular to the surface of the slab. The microwave oven operates at a frequency of 2.45 GHz and produces an electric field intensity with amplitude 500 V/m inside the dielectric, calculate the displacement current density in the dielectric.

Solution: The electric field inside the slab can be given by

$$\mathbf{E} = \hat{n} E_0 \sin(\omega t)\,[\mathrm{V/m}] \Rightarrow \mathbf{D} = \hat{n}\epsilon_0\epsilon_r E_0 \sin(\omega t)\,\left[\mathrm{C/m^2}\right]$$

where $\hat{n}$ is the direction perpendicular to the slab surface and E_0 is the field amplitude.

The displacement current density is given by

$$\mathbf{J_d} = \frac{\partial \mathbf{D}}{\partial t} = \hat{n}\epsilon_0\epsilon_r \omega E_0 \cos(\omega t)\,\left[\mathrm{A/m^2}\right]$$

Given $\epsilon_r = 2$, $E_0 = 500$ V/m, $\omega = 2\pi \times 2.45 \times 10^9$ rad/sec,

$$\therefore \mathbf{J_d} = 136.3 \cos(4.9 \times 10^9 \pi t)\,\left[\mathrm{A/m^2}\right]$$

3. a) Show that the ratio of the amplitudes of the conduction current density and the displacement current density is $\sigma/\omega\epsilon$ for the applied field $\mathbf{E} = \mathbf{E}_m cos\omega t$. Assume $\mu = \mu_0$.

   b) Assuming that sea water has $\mu = \mu_0$, $\epsilon = 81\epsilon_0$, $\sigma = 20$ S/m, determine the frequency at which the conduction current density is ten times the displacement current density in magnitude.

Solution:

a) The displacement vector

$$\mathbf{D} = \epsilon\mathbf{E} = \epsilon\mathbf{E}_m cos\omega t$$

and the displacement current density is

$$\mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t} = -\omega\epsilon\mathbf{E}_m sin\omega t = \omega\epsilon\mathbf{E}_m cos(\omega t + \pi/2)$$

The conduction current density is

$$\mathbf{J}_c = \sigma\mathbf{E} = \sigma\mathbf{E}_m cos\omega t$$

Using these two results we find

$$\frac{|\mathbf{J}_c|}{|\mathbf{J}_d|} = \frac{\sigma\mathbf{E}_m}{\omega\epsilon\mathbf{E}_m} = \frac{\sigma}{\omega\epsilon}.$$

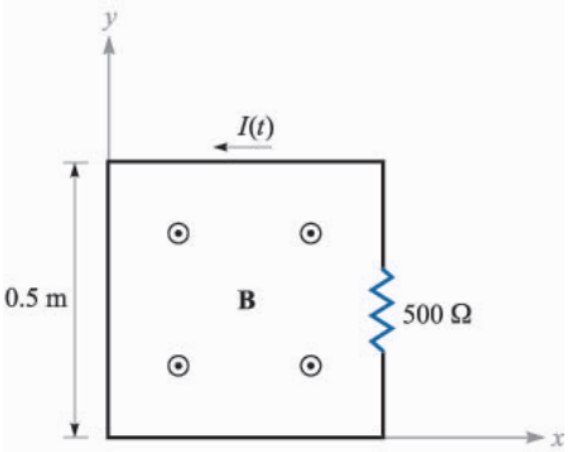
b)

$$\frac{J_c}{J_d} = \frac{\sigma}{\omega\epsilon} = 10 \tag{1}$$

$$\implies \omega = 2\pi f = \frac{\sigma}{10\epsilon} = \frac{20}{10 \times 81\epsilon_0}$$

Solving, we get $f = 0.44$ GHz

4. A perfectly conducting filament containing a small 500Ω resistor is formed into a square, as illustrated by Fig. **??**. Find $I(t)$ if (a) $\mathbf{B} = 0.3\ \cos(120\pi t - 30°)\mathbf{a}_z$ T; (b) $\mathbf{B} = 0.4\ \cos[\pi(ct - y)]\mathbf{a}_z\ \mu$T, where $c = 3 \times 10^8$ m/s.



Solution:

Area of the square loop = $0.5 \times 0.5 = 0.25$ m^2

$\texttt{emf} = -\frac{d\phi_B}{dt}$, R = 500Ω and $d\mathbf{S} = dxdy\hat{a_z}$

(a) $\phi_B = \int_S \mathbf{B} \cdot d\mathbf{S} = \int_S 0.3cos(120\pi t - 30°)\mathbf{a}_z \cdot d\mathbf{S} = 0.3cos(120\pi t - 30°) \cdot (0.25)$

$\implies \phi_B = 0.075$ cos (120πt - 30°) [Wb]

$\texttt{emf} = -\frac{d\phi_B}{dt} = 28.27\sin(120\pi t - 30^o)$ [V]

$\therefore I(t) = \frac{emf}{R} = 0.057\sin(120\pi t - 30°)$ [A]

(b) $\phi_B = \int_S \mathbf{B} \cdot \mathbf{dS} = \int_S 0.4\cos(\pi(ct - y)) \cdot 10^{-6}\mathbf{a}_z \cdot \mathbf{dS}$

$= \int_0^{0.5}\int_0^{0.5} 0.4\cos(\pi ct - \pi y) \cdot 10^{-6}dxdy$

$\implies \phi_B = 0.5 \times 0.4 \times 10^{-6}[\sin(\pi ct - \pi y)]_0^{0.5} \times \frac{-1}{\pi}$

$= \frac{-0.2\times10^{-6}}{\pi}[sin(\pi ct - 0.5\pi) - sin(\pi ct)]$ [Wb]

$\texttt{emf} = -\frac{d\phi_B}{dt} = -0.2c \times 10^{-6}[\cos(\pi ct - 0.5\pi) - \cos(\pi ct)]$ [V]

$\therefore I(t) = \frac{emf}{R} = 0.12[\cos(\pi ct - 0.5\pi) - \cos\pi ct]$ [A]

5. Consider a point on the surface of a perfect conductor. The electric field intensity at that point is $\mathbf{E} = (500\hat{x} - 300\hat{y} + 600\hat{z})\cos 10^7 t$ and medium surrounding the conductor is characterized by $\mu_r = 5$ and $\epsilon_r = 10$ and $\sigma = 0$.
   (a) Find a unit vector normal to the conductor at that point of the conductor surface.
   (b) Find the instantaneous surface charge density at the point.

Solution: (a) Since we have a perfect conductor, Electric field at any point will be normal and directed out of the surface. So it will be given by outward normal electric field vector at that point.

$$\implies \hat{n} = \frac{\mathbf{E}(t=0)}{|E(t=0)|}$$

$$\implies \hat{n} = \frac{5\hat{x} - 3\hat{y} + 6\hat{z}}{\sqrt{(5^2 + 9 + 36)}} = 0.60\hat{x} - 0.36\hat{y} + 0.72\hat{z}$$

(b) Since it is a metallic conductor, no field will exist inside the metal. Hence using boundary conditions, surface charge density is given by

$$(\mathbf{D_1}.\hat{n} - \mathbf{D_2}.\hat{n}) = \rho_s$$

Here D_2 will be zero as it is zero within the metallic conductor $\therefore$ the instantaneous surface charge density will be

$$(\mathbf{D_1}.\hat{n}) = \rho_s$$

$\implies \rho_s = 10\epsilon_0(500\hat{x} - 300\hat{y} + 600\hat{z})(\cos 10^7 t).(0.60\hat{x} - 0.36\hat{y} + 0.72\hat{z}) = 74\cos 10^7 t$ [nC/m^2]

6. A parallel-plate capacitor with plate area of 5 cm^2 and plate separation of 3 mm has a voltage

$$V(t) = 50\sin(10^3 t) \text{ V}$$

applied to its plates. Calculate the displacement current assuming $\varepsilon = 2\varepsilon_0$.

Solution:

The electric flux density is

$$D = \varepsilon E = \varepsilon \frac{V}{d}.$$

The displacement current density is

$$J_d = \frac{\partial D}{\partial t} = \frac{\varepsilon}{d}\frac{dV}{dt}.$$

Hence, the total displacement current is

$$I_d = J_d \cdot S = \frac{\varepsilon S}{d}\frac{dV}{dt} = C\frac{dV}{dt},$$

which is the same as the conduction current:

$$I_c = \frac{dQ}{dt} = S\frac{dD}{dt} = \varepsilon S\frac{dE}{dt} = \frac{\varepsilon S}{d}\frac{dV}{dt} = C\frac{dV}{dt}.$$

Substituting numerical values:

$$I_d = 2 \cdot \frac{10^{-9}}{36\pi} \cdot \frac{5 \times 10^{-4}}{3 \times 10^{-3}} \cdot 10^3 \times 50\cos(10^3 t)$$

$$I_d = 147.4\cos(10^3 t) \text{ nA}.$$

7. Given three of Maxwell's equations in vacuum as $\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H}$, $\nabla \times \mathbf{H} = j\omega\epsilon\mathbf{E}$ and $\nabla \cdot \mathbf{E} = 0$, how can we derive a single equation for the $\mathbf{E}$ field? Don't solve the equation, just derive it.

Solution:

Using the equation

$$\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H} \tag{2}$$

and the curl operation, we get

$$\nabla \times (\nabla \times \mathbf{E}) = -j\omega\mu(\nabla \times \mathbf{H})$$

Now, using the following identity on (2),

$$\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2\mathbf{A}.$$

we get,

$$\nabla(\nabla \cdot \mathbf{E}) - \nabla^2\mathbf{E} = -j\omega\mu(j\omega\epsilon\mathbf{E})$$

Finally,

$$\nabla^2\mathbf{E} = \gamma^2\mathbf{E}, \quad \gamma^2 = -\omega^2\mu\epsilon$$

8. In a non-magnetic material ($\mu_r = 1$) with dielectric constant $\epsilon_r = 4$, the electric field is given by $\mathbf{E}(t) = 20\sin(10^8 t - \beta z)\hat{y}$. Calculate the propagation constant β and the magnetic field $\mathbf{H}(t)$.

Solution: We know that

$$\begin{aligned}\beta &= \omega\sqrt{\mu\epsilon} \\ &= 10^8\sqrt{\mu_0 \times 4\epsilon_0} = \frac{10^8(2)}{3\times 10^8} = \frac{2}{3}\end{aligned}$$

Now converting the electric field into phasor form : $\mathbf{E}(t) = Im(\mathbf{E}e^{j\omega t}) \Rightarrow \mathbf{E} = 20e^{-j\beta z}\hat{y}$. From the Maxwell's equation

$$\begin{aligned}\nabla \times \mathbf{E} &= -j\omega\mu\mathbf{H} \\ \mathbf{H} &= \frac{\nabla \times \mathbf{E}}{-j\omega\mu} \\ &= \frac{1}{-j\omega\mu}\left[-\frac{\partial E_y}{\partial z}\hat{x}\right] \\ &= -\frac{20\beta}{\omega\mu}e^{-j\beta z}\hat{x}\end{aligned}$$

Substituting the value of β, we get

$$\mathbf{H} = -\frac{1}{3\pi}e^{-j\beta z}\hat{x}$$

So the sinusoidal form of magnetic field is given by

$$\begin{aligned}\mathbf{H}(t) &= Im(\mathbf{H}_s e^{j\omega t}) \\ &= -\frac{1}{3\pi}\sin(10^8 t - \beta z)\hat{x}\end{aligned}$$

9. A 50 V voltage generator at 20 MHz is connected to the plates of an air-dielectric parallel-plate capacitor with plate area 2.8 cm^2 and separation 0.2 mm. Find the *maximum* displacement current density and the displacement current.

Solution: For a capacitor with sinusoidal excitation,

$$J_d = j\omega\mathbf{D}, \qquad J_{d\max} = \omega\epsilon E = \omega\epsilon\frac{V}{d}.$$

With air, $\epsilon = \epsilon_0 = \dfrac{10^{-9}}{36\pi}$ F/m, $\omega = 2\pi f = 2\pi \times 20 \times 10^6$ rad/s, $V = 50$ V, and $d = 0.2\,\text{mm} = 2.0 \times 10^{-4}$ m:

$$\boxed{J_{d\max} = \omega\epsilon_0\frac{V}{d} = \frac{10^{-9}}{36\pi}(2\pi \times 20 \times 10^6)\frac{50}{0.2\times 10^{-3}} \approx 277.8\,\text{A/m}^2.}$$

The plate area is $S = 2.8\,\text{cm}^2 = 2.8 \times 10^{-4}\,\text{m}^2$, hence the maximum displacement current is

$$\boxed{I_d^{\max} = J_{d\max}S = 277.8 \times 2.8 \times 10^{-4} \approx 7.78 \times 10^{-2}\,\text{A} = 77.78\,\text{mA}.}$$

10. The ratio of conduction to displacement current density is

$$\frac{J_c}{J_d} = \frac{\sigma E}{\omega \epsilon E} = \frac{\sigma}{\omega \epsilon}.$$

Compute this ratio at 1 GHz for:

(a) distilled water: $\mu = \mu_0$, $\epsilon = 81\,\epsilon_0$, $\sigma = 2 \times 10^{-3}\,\mathrm{S/m}$,
(b) sea water: $\mu = \mu_0$, $\epsilon = 81\,\epsilon_0$, $\sigma = 25\,\mathrm{S/m}$,
(c) limestone: $\mu = \mu_0$, $\epsilon = 5\,\epsilon_0$, $\sigma = 2 \times 10^{-4}\,\mathrm{S/m}$.

Solution: Let $f = 1\,\mathrm{GHz}$ so $\omega = 2\pi f = 2\pi \times 10^9\,\mathrm{rad/s}$ and $\epsilon_0 = \dfrac{10^{-9}}{36\pi}\,\mathrm{F/m}$.

(a) **Distilled water** ($\epsilon = 81\,\epsilon_0$, $\sigma = 2 \times 10^{-3}\,\mathrm{S/m}$):

$$\boxed{\frac{J_c}{J_d} = \frac{\sigma}{\omega \epsilon} = \frac{2 \times 10^{-3}}{(2\pi \times 10^9)\,(81)\left(\dfrac{10^{-9}}{36\pi}\right)} \approx 4.4 \times 10^{-4}.}$$

(b) **Sea water** ($\epsilon = 81\,\epsilon_0$, $\sigma = 25\,\mathrm{S/m}$):

$$\boxed{\frac{J_c}{J_d} = \frac{25}{(2\pi \times 10^9)\,(81)\left(\dfrac{10^{-9}}{36\pi}\right)} \approx 5.555.}$$

(c) **Limestone** ($\epsilon = 5\,\epsilon_0$, $\sigma = 2 \times 10^{-4}\,\mathrm{S/m}$):

$$\boxed{\frac{J_c}{J_d} = \frac{2 \times 10^{-4}}{(2\pi \times 10^9)\,(5)\left(\dfrac{10^{-9}}{36\pi}\right)} \approx 7.2 \times 10^{-4}.}$$

11. In a certain region,

$$\mathbf{J} = (2y\,\hat{\mathbf{a}}_x + xz\,\hat{\mathbf{a}}_y + z^3\,\hat{\mathbf{a}}_z)\,\sin(10^4 t)\ \mathrm{A/m^2}.$$

Find $\rho_v(x, y, z, t)$ if $\rho_v(x, y, 0, t) = 0$ holds.

Solution:

$$\nabla \cdot \mathbf{J} = (0 + 0 + 3z^2)\sin 10^4 t \ = \ -\frac{\partial \rho_v}{\partial t}$$

$$\rho_v = \int \nabla \cdot \mathbf{J}\, dt = \int 3z^2 \sin 10^4 t\, dt = -\frac{3z^2}{10^4}\sin 10^4 t + C_0$$

If $\rho_v\big|_{z=0} = 0$, then $C_0 = 0$ and

$$\boxed{\rho_v = -0.3\,z^2 \sin 10^4 t\ \mathrm{mC/m^3}}$$